



Project Proposal

Smart Food Waste Reduction & Redistribution System

Course Name:	Software Engineering Design Capstone Project
Course Code:	SE 331 (F1)
Supervisor:	Rahat Uddin Azad, Lecturer, DIU, SWE
University:	Daffodil International University

Group Members

ID	Name
0242310005341219	Md.Mahbub Alam
0242310005341015	Tanvir Ahmed
0242310005341171	Md Muhtasim Fuad Rahat

1. Introduction

Food wastage is a serious social and economic problem where large amounts of edible food are discarded while many people remain hungry. Restaurants, hotels, events, and households often have surplus food that goes to waste due to lack of coordination with organizations or people who need it. This project proposes a web-based system that connects food providers with NGOs and volunteers to redistribute surplus food efficiently.

The proposed system aims to reduce food waste, improve coordination, and ensure that surplus food reaches people in need through a centralized digital platform.

1.1 Background Overview

This project falls under the smart system and social welfare application domain.

Currently in Bangladesh, food redistribution mostly depends on manual communication through phone calls, Facebook groups, or personal contacts. NGOs and volunteers often struggle to find available surplus food quickly before it expires. Similarly, food providers cannot always find receivers in time.

Existing initiatives work on food redistribution but:

- Coordination is mostly manual
- No centralized digital platform
- No real-time notifications
- Limited tracking and reporting

A web-based smart system can improve efficiency, transparency, and coordination.

1.2 Problem Statement

Large amounts of edible food are wasted daily due to poor coordination between food providers and receivers. Providers often have surplus food but cannot find receivers quickly. NGOs and volunteers also face difficulty locating available food in time.

Current systems are:

- Manual and slow
- Not centralized
- Lack real-time updates
- Difficult to track

As a result, food gets wasted even when people need it. A realistic and feasible software-based solution is required to connect providers and receivers through a centralized platform for faster communication and tracking.

1.3 Objectives

General Objective

To develop a web-based system that connects surplus food providers with NGOs and volunteers to reduce food waste and improve redistribution.

Specific Objectives

- Reduce food wastage
- Enable real-time food redistribution
- Improve transparency and tracking
- Provide centralized coordination
- Generate reports and analytics
- Ensure verified providers and NGOs
- Monitor actual food distribution

1.4 Scope

In-Scope Features

- User registration and login
- Legal document verification of providers/NGOs by admin
- Food posting by providers
- Viewing available food
- Food claiming by receivers
- Notification system (within 5 km radius)
- Pickup confirmation
- NGO distribution proof upload (photo/report)
- Monitoring and blocking system if misuse occurs
- Admin dashboard
- Reports and tracking

Out-of-Scope Features

- Transportation logistics
- Payment systems
- Advanced AI prediction
- Mobile application (future work)

The system focuses only on coordination and tracking of surplus food.

1.5 Stakeholders

Food Providers

Restaurants, hotels, and households who donate surplus food.

Interest: Reduce waste and donate easily.

NGOs / Volunteers(Distributor)

Receive notifications and collect food.

Interest: Quickly find available food and distribute.

System Administrator

Manages users, verifies documents, monitors distribution, and generates reports.

Interest: Maintain system performance, transparency, and security.

1.6 Proposed Solution

The proposed system is a web platform where providers can post surplus food details such as quantity and expiry time. Receivers can view and claim food. The system prioritizes food based on expiry time and distance and sends notifications to nearby receivers within a 5 km radius.

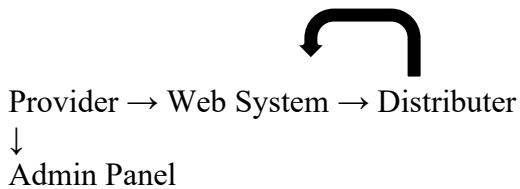
Admin verifies legal documents of providers and NGOs. After food distribution, NGOs must upload proof (photo/report). If they fail to upload proof twice, their account will be blocked.

After pickup, the receiver confirms collection and the system stores records for reporting.

Workflow

1. Provider posts food
2. System notifies receivers within 5 km
3. Receiver claims food
4. Provider confirms
5. Receiver collects food
6. NGO uploads distribution proof
7. Admin monitors
8. System records transaction

Simple Overview Diagram



2. System Requirements

Functional Requirements

- The system shall allow users to register and login.
- The admin shall verify legal documents of providers/NGOs.
- The system shall allow providers to post surplus food.
- The system shall display available food to receivers.
- The system shall notify nearby receivers within 5 km.
- The system shall allow receivers to claim food.
- The system shall allow providers to confirm requests.
- The system shall store transaction records.
- The system shall allow NGOs to upload distribution proof.
- The system shall block users after repeated violations.
- The system shall allow admin to generate reports.

Non-Functional Requirements

- Response time under 2 seconds
- Secure authentication and validation
- User-friendly interface
- 24/7 availability
- Data privacy protection

3. Tools and Technologies

Frontend: HTML, CSS, React, Tailwind CSS, JavaScript

Backend: Node.js, Express

Database: MongoDB

Authentication: JWT

Tools: GitHub, VS Code

4. Project Timeline and Work Plan

Phase	Task	Duration
Week 1–2	Requirement analysis & design	2 weeks
Week 3–5	Backend development	3 weeks
Week 6–7	Frontend development	2 weeks
Week 8	Integration	1 week
Week 9	Testing	1 week
Week 10	Documentation	1 week
Week 11–12	Final improvements	2 weeks

Team Work Division

- Member 1: Backend & database
- Member 2: Frontend UI
- Member 3: Testing & documentation

5. Optional Sections

Existing System / Related Initiatives in Bangladesh

Some NGOs and initiatives in Bangladesh already work on food redistribution:

Hunger Bridge Bangladesh

Collects surplus food and distributes to people in need.

Pari Foundation

Collects leftover food from events and distributes to underprivileged communities.

Zero Hunger Bangladesh (ULAB)

Student initiative for food redistribution.

BRAC Food Programs

Work on reducing food loss and supporting communities.

Foodi App (Food Rescue Feature)

Allows restaurants to sell surplus food at discounted prices.

Gap in Existing Systems

- Mostly manual coordination
- No centralized platform
- No automated notification
- Limited tracking

The proposed system provides a centralized and automated digital solution.

System Architecture

- Frontend (Web UI)
- Backend (API & logic)
- Database
- Notification module

Ethical, Legal, and Social Considerations

- Secure user data
- Legal document verification
- Food safety disclaimer
- Positive social impact

Future Work

- Mobile app
- ML-based prediction
- Route optimization